

```
url <- "https://raw.githubusercontent.com/roualdes/data/master/email.csv"
suppressMessages(email <- read.csv(url))
head(email)
```

```
##      spam to_multiple from cc sent_email      time image attach dollar
## 1      0              0      1 0          0 2011-12-31 22:16:41      0      0      0
## 2      0              0      1 0          0 2011-12-31 23:03:59      0      0      0
## 3      0              0      1 0          0 2012-01-01 08:00:32      0      0      4
## 4      0              0      1 0          0 2012-01-01 01:09:49      0      0      0
## 5      0              0      1 0          0 2012-01-01 02:00:01      0      0      0
## 6      0              0      1 0          0 2012-01-01 02:04:46      0      0      0
##      winner inherit viagra password num_char line_breaks format re_subj
## 1      no          0          0          0 11.370          202      1      0
## 2      no          0          0          0 10.504          202      1      0
## 3      no          1          0          0  7.773          192      1      0
## 4      no          0          0          0 13.256          255      1      0
## 5      no          0          0          2  1.231           29      0      0
## 6      no          0          0          2  1.091           25      0      0
##      exclaim_subj urgent_subj exclaim_mess number
## 1              0              0              0    big
## 2              0              0              1  small
## 3              0              0              6  small
## 4              0              0             48  small
## 5              0              0              1   none
## 6              0              0              1   none
```

```
xbar <- mean(email$num_char)
s <- sd(email$num_char)
n <- length(email$num_char)
alpha <- 0.05
se <- s / sqrt(n)
t.star <- qt(1 - alpha, n - 1)
ME <- se * t.star
c(-Inf, xbar + ME)
```

```
## [1]      -Inf 11.09139
```

Lightbulb

```
mu <- 12000
n <- 64
xbar <- 11297
s <- 2418
alpha <- (1 - 0.87)
se <- s / sqrt(n)
tstar <- qt(1 - alpha / 2, n - 1)
me <- se * tstar
c(xbar - me, xbar + me)
```

```
## [1] 10833.3 11760.7
```